BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

SECOND SEMESTER 2025-26
COURSE HANDOUT

Date: 05.01.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **EEE/INSTR/ECE F242**
Course Title : **Control Systems**
Instructor-in-Charge : **Dr. Puneet Mishra**
Instructor(s) : **Dr. Puneet Mishra, Dr. Prashant Kumar**
Tutorial/Practical Instructors: Dr. Hari Om Bansal, Dr. Prashant Kumar, Dr. Aditya R Gautam, Dr. Pawan Ajmera, Dr. Puneet Mishra

1. Course Description: Modeling and classification of dynamical systems, Properties and advantages of feedback systems, time-domain analysis, frequency-domain analysis, stability and performance analysis, State space analysis, controller design.

2. Scope and Objective of the Course: Feedback or automatic control is an essential feature of numerous industrial processes, scientific instruments and even commercial, social and management situations. A thorough understanding of the elementary principles of this all-embracing technology is of great relevance for all engineers and scientists in general and Electrical & Electronics engineers in particular. This course tries to bring out the basic principles of Feedback Control Systems.

3. Learning Outcomes:

- *Modeling & Representation:* By the end of the course, students will be able to derive mathematical models and transfer functions for electrical, mechanical, and electromechanical systems, and represent them using block diagrams and signal flow graphs.
- *System dynamic and steady state behavior & its analysis:* Students will be able to analyze closed-loop control systems for stability and transient response under different excitations and evaluate the impact of controllers such as P, PI or PID on system performance.
- *Graphical & Frequency Techniques:* Students will be able to construct root locus and frequency response plots (Bode, Polar and Nyquist plots) to assess system stability and performance and use these tools for design insights.
- *Modern Control Concepts:* Students will be able to apply model order reduction methods, distinguish between discrete-time and continuous-time system representations, and utilize state-space modeling as a modern framework for analyzing dynamical systems.

4. Text Books: I. J. Nagrath and M. Gopal, Control Systems Engineering, New Age Publishers

5. Reference Books:

(i) Kuo, B. C., and Golnaraghi, F., Automatic Control Systems, John Wiley & Sons, 8th Ed, 2003.

(ii) K. Ogata, Modern Control Systems, Pearson Education, 4th Ed., 2002

(iii) Drof, R. C., and Bishop, R. H., Modern Control Systems, Addison Wesley, 7th Ed, 1995.

(iv) Nise's control systems engineering by Nise, Norman S, 7th ed, Wiley, 2015

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1. Introduction (Lec1-2)	Introduction; Different types of systems and signals; Various terminologies; Open loop and closed loop control; A brief history; Examples from various fields	Ch.1 of Textbook, Lecture Notes	General understanding of the concept of control systems; familiarization with various terms and their significance; familiarization with various application domains.
2. Mathematical Modelling (Lec 3-8)	Introduction to mathematical modelling; Modelling of electrical and mechanical systems; Block diagram representation and simplification; Properties of feedback control	Ch. 2 & 3 of Textbook, Lecture Notes	Learning how to derive mathematical models and transfer functions of electrical, mechanical, electromechanical systems; Understanding block diagram & signal flow graph representation and simplification; understanding about properties of closed loop control systems
3. Control Systems Components (Lec 9-10)	DC servomotors; AC servomotor; Stepper motor; Synchro;	Ch. 4 of Textbook, Lecture Notes	Learning about various control systems components and their use; Forming block diagrams and deriving transfer functions
4. Stability analysis (Lec 11-12)	Stability determination using R-H criteria	Ch. 6 of Textbook, Lecture Notes	Learning how to carry out stability analysis of dynamic systems to different excitations
5. Time Domain Analysis (Lec 13-19)	Various test signals and time response of 1st and 2nd order systems to them; Time domain specifications and their expressions for prototype 2nd order systems; Steady state errors; Introduction to PID controller	Ch. 6 &11 of Textbook, Lecture Notes	Learning how to carry out transient response analysis of dynamic systems to different excitations; Understanding the effects of PID type controllers
6. Root Locus Technique (Lec 20-22)	Concept of root locus; Method of drawing root locus plot; Various examples	Ch.7 of Textbook, Lecture Notes	To draw root locus for various systems and therefrom infer information on time response and stability
7. Frequency Domain Analysis-I (Lec 23-28)	Concept of frequency domain and comparison with time domain; Frequency domain specifications; Polar plot; Gain and Phase margins Nyquist stability criterion	Ch. 8 & 9 of Textbook, Lecture Notes	To learn how to obtain various frequency response plots of systems and how to use them for stability and performance analysis

8. Frequency Domain Analysis - II (Lec 29-32)	Bode plot; Transfer function identification from Bode plot; non-minimum phase systems; Lead, Lag controller design	Ch. 8, 9 &10 of Textbook, Lecture Notes	Do
9. Introduction to model order reduction (Lec 33-34)	Pade approximation and Routh Approximation methods	Lecture Notes	To learn how to reduce the order of a system using different methods.
10. Introduction to Discrete-time Systems (Lec 35-37)	Representation of Discrete Time systems; Sampling and reconstruction. ; Stability analysis of Discrete Time systems	Ch.11 of Textbook, Lecture Notes	Learning about mathematical representation of Discrete Time systems vis-a-vis Continuous Time systems
11. Introduction to State Space Modelling and Concluding Remarks (Lec 38-40)	Introduction to State Space modelling approach; Examples; Comparison with Transfer Function approach; Interconversions between transfer functions and state space models; A brief overview of further topics on Control Systems	Ch.12 of Textbook (initial parts), Lecture Notes	Learning about State Space representation of dynamic systems vis-a-vis transfer function representation and a flavour of more advanced aspects of the area of control engineering

7. Evaluation Scheme:

Component	**Duration**	**Weightage (200 M)**	**Date & Time**	**Nature of component (Close Book / Open Book)**
Mid-Semester Test	1.5 Hrs.	30% (60 M)	10/03 AN1	OB
Comprehensive Examination	3 Hrs.	45% (90 M)	06/05 FN	CB
Quizzes: (Best 5 out of 6 Quizzes) - **Announced**	10 Min.	15% (30 M)	During Tutorial Hours	CB
Assignment	Continuous	10 % (20 M)	TBA	OB

8. Chamber Consultation Hour: To be announced in the class

9. Notices: All notices will be displayed on NALANDA

10. Make-up Policy: No make-up will be given for quizzes and assignments; however, make-up exam applications will be approved ONLY in extremely genuine cases for other components. In such cases, the student must provide sufficient proof or obtain prior permission from the IC.

11. Minimum performance requirement for a valid grade: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

Instructor-in-charge
Course No. ECE/ EEE/INSTR F242